INVESTIGATION AND REDUCTION OF RADIATION HEAT LOSSES FROM A ROTARY KILN BY USING LOW EMISSIVITY PAINT

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ABSTRACT:

Rotary kilns are thermal processing tools that are used for processing solid materials at extremely high temperature. In cement industries, calcination process takes place inside a rotary kiln which is a very energy intensive process. Global focus is shifting towards energy efficiency and it requires exploration of different means which can help in achieving this goal. This paper investigates saving of radiation heat losses from kiln surface by applying low emissivity paint to kiln surface and simulate the physical behavior of kiln inside using ANSYS. This paper also explores the need for a critical emissivity value as decreasing the surface emissivity increases the surface temperature and thus therefore it is necessary to evaluate the exact effect of decreasing the emissivity of the kiln surface on its temperature. A sharp decrease in surface emissivity will induce extremely high thermal stresses that can significantly reduce the structural integrity of the kiln. A detailed heat transfer analysis is conducted to quantify the effect of surface emissivity on surface temperature of the kiln. We have found that for every 15°C change in surface temperature the total heat losses changes by almost 10%-12%. Our result implies the use of a paint with 0.6 emissivity that will ensure a safe increase in surface temperature to avoid causing structural damage. The energy saving in this instance will be 250KW which when converted in terms of fuel used with a net calorific value (25958 KJ/Kg), the daily reduction in fuel consumption will be 832 kg of coal.

Keywords: kiln, radiations, convection, emissivity, ANSYS, heat loss

Introduction:

Rotary kiln are advance thermal processing units that are used by various industries to process solid materials at extremely highly temperature. In cement industries, calcination process takes place inside the rotary kiln. This process requires temperature greater than 1000 ^{o}C [1]. Because of such a high temperature on the inside, the kiln surface gets heated and its surface temperature can reach up to a value of 300 ^{o}C. For a 4.4 m diameter kiln having length 52m whose average surface temperature is 300°C, the total heat losses from its surface are 5173 KW, 30% of the total heat loses occurs due to convection while 70% losses occur due to thermal radiations.

For complete analysis of the rotary kiln, the internal wall and gases temperature are needed. The temperature inside the kiln can reach up to 2000 Celsius[2] and that is why it is not possible to place any sensor inside the rotary kiln that can give us the readings we need. The operator controls the whole process based on the condition of the clinker at inlet and outlet and then changes the parameters that are in operator control to get the desired inside conditions [3]. In order to find the internal temperature profile of the wall as well as the combustible gases, we have used a computational fluid dynamics (CFD) program. CFD will help us predict the gas and wall temperature at a given mass flow rate of fuel (Coal) and air which are used in the factory. The results obtained from CFD will then be used to find the heat loss to the atmosphere which is a combination of radiation heat loss and convection heat loss.

Emissivity is a surface radiative property that determines how efficiently the surface can emit radiation. The emissivity of any surface can assume different values[4], depending upon emission of a particular wavelength, emission in a particular direction, emission in integrated averages over wavelength, emission in integrated averages over direction. A diffuse surface is one for which emitted radiation are independent of direction in which they emit those radiations. Thus, we assume that kiln surface to be diffuse. We are interested in emissions of all wavelength radiation; therefore, the values of kiln surface emissivity correspond to third case. I-e, Normal emission from the surface over all wavelengths, thus the total hemispherical emissivity can be approximated as normal emissivity (θ=0, or perpendicular to surface).

Radiation barrier coating or low emissivity paints are formulated to reduce radiant heat transfer by reducing emissivity of a surface. These paints are widely applicable for energy saving and aircrafts stealth technology due to their low cost

and simple construction technology. Low emissivity paints applied to concrete walls in houses will keep heat inside in winter and outside in summers. The emissivity of concrete is 0.94 without the application of paints and will be 0.145 if low emissivity paint low-mit2 a product of solec solar energy corporation is used.

Low emissivity coating is usually composed of polymer binder and pigments such as aluminum or silver. These coating have good structural stability and low emissivity performance below 300 °C, but when the temperature is above 300 °C, their Structural integrity is reduced due to high temperature. Study shows that paint that is composed of Epoxy modified silicon as a binder can maintain it properties up to 600 ºC. Emissivity of the specific paint depends upon the Ratio of binder to pigment and the required paint of 0.6 emissivity can be achieved. Coating with a P/B ratio of 1.0 can maintain its integrity up to 600 °C and its lowest emissivity value is 0.27 at 600 °C. Emissivity decreases with increase in pigments fraction in the composition, but increases the probability of oxidation at 600ºC and above, and with the oxidation of surface, its emissivity increases.

The ratio of total emissivity and normal emissivity has different ranges for conductors and non-conductors. Approximating total emissivity from normal emissivity can have ±5% error for conductors and ±3% error for the non-conductors. Thus, the approximation is appropriate. Also, the emissivity of surface depends upon nature of the surface. For any material, its surface nature is affected by the method of fabrication, thermal cycling and chemical reaction with the environment [4]. Since the material for the kiln shell is generally steel, mild steel (or) stainless steel and it is exposed to heavy oxidation therefore assuming an initial surface emissivity of 0.9 for dark brown, heavily oxidized kiln surface.

ANSYS Fluent Use Finite volume method in which the whole geometry or domain is discretized into finite number of cells or volumes. In our analysis, we are using data from a real kiln. The source of the data is Cherat Cement Factory. We collected the data on our visit to the factory. There are three operational rotary kilns in Cherat Cement factory. We will be performing our analysis on Line 2 in the factory.

Methodology:

In this study, we developed a technique to minimize the radiation heat losses from the kiln surface by applying low emissivity paint. Applying low emissivity paint to the kiln surface can significantly reduce radiation heat losses[5]. However, the effect of decreasing surface emissivity must be carefully evaluated because decrease in surface emissivity will increase kiln shell temperature. Too much decrease in surface emissivity will increase surface temperature to a point where thermal stresses in kiln shell will become high enough to collapse the entire kiln structure. The steps taken during our study are given in the flow chart.

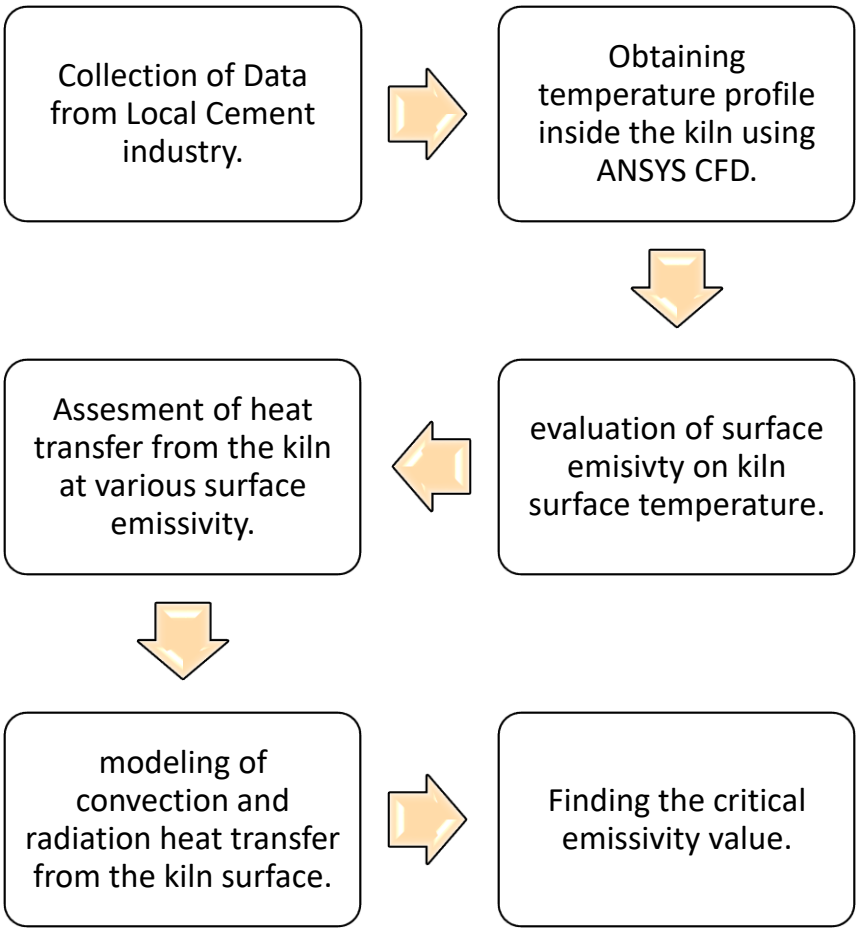


Figure 1: Flowchart of different steps taken during our study to asses energy efficacy of low emissivity paint for rotary kiln.

To model the rotary kiln in ANSYS CFD, certain assumptions were made which are as follows.

- We assume the kiln to be a simple cylindrical structure.
- Steady state conditions.
- We do not consider effect of gravity.
- The flow is incompressible.

The geometry is made using ANSYS Space Claim. The geometry is modeled using real kiln data from the factory. The primary air enters the kiln in three concentric pipes along the fuel inlet pipe. The geometry is set such that the burner tip is 1 meter inside the kiln while the secondary inlet is at the start of the rotary kiln.

Table 1: Shows kiln geometry and burner specification from kohat cement factory.

Kiln Geometry		
Geometric specifications	**Quantity**	**Units**
Length	52000	mm
Inner Diameter	4000	mm
Outer Diameter	4400	mm
Burner Specifications		
Burner type	Sino flame full-power coal burner	
Diameter of axial air inlet	395	mm
Diameter of fuel inlet	355	mm
Diameter of swirl flow inlet	450	mm
Diameter of secondary air inlet	4000	mm
Diameter of the outlet	4000	mm

Meshing is the discretization of the geometry into finer elements which are then used to calculate the equations at each intersection of these elements. The finer the mesh, the greater is the accuracy but the increased accuracy incurs a heavy computational cost and for that reason a balance is needed [6]. In order to achieve the desired accuracy, we refined the mesh in such a way that the area near the burner tip has a finer mesh and element size while the area away from tip has course mesh size. The rationale behind the refinement is that we restrict the mesh elements to a level which is not computationally too expensive but give accurate results at the same time. The area where combustion takes place is very sensitive to change and requires a very fine mesh size. From the figure, one can see that the mesh is fine in the combustion zone. The shape of the cell we used here is a triangle, as it requires less computation power [6] as shown in figure 2.

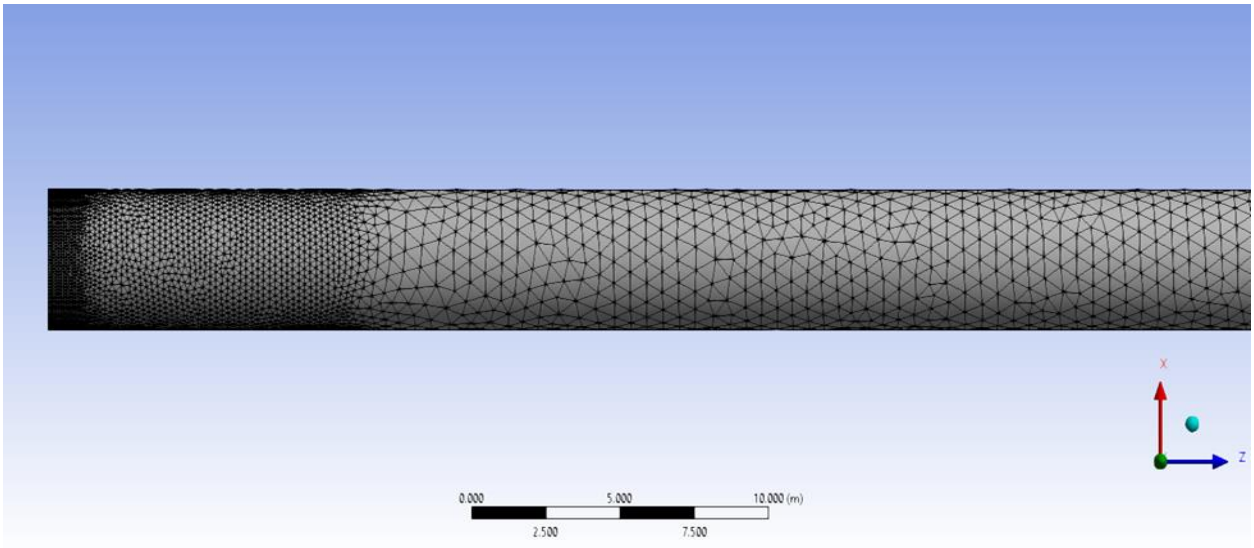


Figure 2: shows how the meshing is done which is refined near the burner tip and coarse as we move along horizontally inside the rotary kiln.

To carry out our analysis we have used a pressure-based solver in ANSYS fluent because we are assuming that the gases are incompressible [7]. We have also neglected the effects of gravity in our modeling. After checking the quality of the mesh, turn on the energy model. This model ensures that the energy has conserved all the time during computing and solution of the kiln [6]. The standard - model is based on model transport equations for the turbulence kinetic energy (k) and its dissipation rate (ε). Since the strengths and weaknesses of the standard – model are known, its performance is improved with some modifications in it. Two of these variants are available in ANSYS Fluent: the RNG - model and the realizable – model. As the flow inside the kiln is turbulent [7], for that we will use RNG k-epsilon model.

For radiation, we used the p1 radiation model, which is suitable for combustion analysis because it considers the scattering effect of radiation. It was important to model radiation to ensure that proper temperature distribution across the entire kiln This model takes into effect the absorption and scattering effects of the gases present and that is why the heat distributes through radiation inside the whole kiln. Therefore, using the p1 radiation model, we get a more distributed temperature profile inside the rotary kiln and that is what previous literature on the subject recommends [8] SPECIES (non-premixed model)

Now in order to assign fuel we go to the species section and use a non-premixed model as shown in figure 2-6, because no mixing of fuel and air occurs prior to combustion [7]. We then make changes in the fuel section based on our needs.in this analysis we are using sub bituminous coal so we make changes accordingly. The normalized values are such that the sum of all the values is equal to one. Based on coal composition (figure 2-7) we get a low calorific value $2.81 \times 10^{07} j / kg$. We then calculate an enthalpy table, which we have used in finding the solution.

After assigning fuel, we go to the cell condition and assign any material to cell where we need it and assign pdf mixture to the flow volume. The boundary conditions used here are derived from a real working kiln (line 2) of the Cherat cement factory. The inlets are mass flow inlets and the exits are pressure-based outlet [9]. The pressure-based outlet assumes standard atmospheric conditions at the outlet of the kiln. The air to fuel ratio used here is 9.3 but with excess of oxygen, it becomes 10.59 kg of air for one kg of coal. This air to fuel ratio is used because we are giving 15 % oxygen in excess to the kiln. The air inlet is divided into primary and secondary inlet. Secondary air takes 90 % of the share while primary takes 10 %. The boundary conditions are given in the table 2.

Table 2: shows the boundary conditions used when simulating the CFD model in ANSYS.

Boundary conditions		
Boundary	**Mass flow rate (kg/s)**	**Temperature (k)**
Total mass flow rate of air	10.59	
Fuel	1	300
axial air	0.1059	300
cool air	0.1059	300
Swirl air	0.8472 (tangential=39, axial=148)	300
Secondary air	9.531	1050

After assigning the boundary conditions, the solution is initialized using hybrid initiation and the solver for all the variables is a second order solver. The solver checks for any recommendation from the software for accurate results and after that we set an iteration value. The solution starts and it is iterated to a set value. The residual lines are given in the figure in the result section which shows the iteration process. We performed the first 250 iterations in first order and then the last 50 in second order spatial discretization.

Now to find the convection heat losses form kiln surface, convection coefficient must be evaluated first for external flow. Three models are presented for external convection coefficient. Depending upon external conditions each model convection coefficient varies, the mixed model convection coefficient is most appropriate because most of the time mixed conditions is favored. When a higher velocity air stream flows over the kiln surface, it results in a forced convection heat transfer mechanism. The forced convection coefficient depends upon the direction of the air flow relative to the kiln. Assuming the forced flow condition, the values of convection coefficient varies along the circumference. To find an average forced convection coefficient over the entire circumference based on average Nusselt number we use equation (1) [4];

$$\overline{N_{u_F}} = 0.3 + \frac{0.62 R_{e_D}^{\frac{1}{2}} P_r^{\frac{1}{3}}}{\left[1+\left(\frac{0.4}{P_r}\right)^{\frac{2}{3}}\right]^{\frac{1}{4}}}\left\{1+\left(\frac{R_{e_D}}{82000}\right)^{\frac{5}{8}}\right\}^{\frac{4}{5}} \qquad (1)$$

we can use equation (1) to calculate the forced convection coefficient, the parameters of which are given in table 3.

Table 3: PARAMETERS FOR FORCED CONVECTION COEFFICIENT.

Quantity	**Symbols**	**Value**	**Unit**
Thermal Conductivity	(K)	0.0368	W/m-K
Outer Diameter of kiln	(D)	4.4	m
Velocity of Air	(v)	4.5	m/s
Kinematic Viscosity	(v)	$31.55 * 10^{-8}$	m^2/s
Reynolds number	(R_{e_D})	627416	-
Prandtl number	(Pr)	0.6865	-
Forced Nusselt number	(N_{u_F})	827.58	-
Convection coefficient	(h_f)	6.9	W/m^2K

Natural convection or free convection, dominates when the velocity of air stream is very low relative to kiln. In free convection buoyancy forces dominates the fluid flow. In a similar manner to find the natural convection coefficient for kiln we approximate nusselt number. In free convection, the Nusselt number is the function of Grashof number and Prandtl number. The Grashof number represents the buoyancy forces relative to viscous force unlike Reynolds number, which represent the inertial forces relative to viscous forces. Now once we approximate nusselt number, we can use equation (1) [4] to approximate the natural convection with the parameters given in table 4.

Table 4:PARAMETERS FOR NATURAL CONVECTION COEFFICIENT.

Quantity	**Symbols**	**Value**	**Unit**
Thermal Conductivity	(K)	0.0368	W/m-K
Outer Diameter of kiln	(D)	4.4	m
Kinematic Viscosity	(v)	$31.55 * 10^{-8}$	m^2/s
Rayleigh number	R_{a_D}	$3.38 * 10^{11}$	-
Prandtl number	Pr	0.6865	-
Forced Nusselt number	N_{u_N}	827.58	-
Grashof number	G_r	6.9	-
Beta	β	0.002257	-
Convection coefficient	h_N	7.22	W/m^2K
Surface Temperature	T_s	300	C
Ambient Temperature	T_∞	40	C

Now to decide which mode of heat transfer dominates, we determine the ratio of Grashof number to the square of Reynolds number. If this ratio is less than unity, than forced convection dominates and the free convection will be neglected in that region. If this ratio is greater than unity, than Natural convection dominates and forced convection will be neglected in that region. If this ratio is equal to unity than then both natural and forced model are prominent and the use of single natural or forced model can lead to greater error in such cases mixed model must be used. The Nusselt number for mixed model depends upon Grashof number, Reynolds number and Prandtl number. To correlate mixed convection heat, transfer for cross flow conditions, we use equation (2) [4];

$$N_u{}^n = N_{u_F}{}^n + N_{u_N}{}^n \qquad (2)$$

For transverse flow over the cylinder the best value of "n" is 4. Thus, from equation (2) we can easily find the Nusselt number from which we can easily find mixed model convection coefficient. Once Nusselt number has been approximated, it is then used in equation (1) to find convection coefficient. From the parameters of forced and natural convection, we have Nusselt number for forced and natural convection. Using these values, we get Nusselt number for the mixed model convection coefficient. Thus using Nusselt number from equation (2) into equation (1) we get a value of 8.4 W/m^2K for mixed convection coefficient.

Assessment of heat losses from Rotary kiln

After modeling heat transfer parameters now, we use them to quantify heat losses from kiln surface. Convection coefficient is used to find convection heat losses while the knowledge of surface emissivity is used to quantify radiation heat losses from kiln surface. Convection heat losses from kiln surface depend upon convection coefficient and surface area of kiln. To quantify the convection heat losses from kiln surface, we use Newton law of cooling [4]. Table 5 below shows data for case study at Cherat cement factory to quantify convection heat losses.

Table 5: Shows parameters for calculation of convection heat transfer from kiln surface (Case study 1 Cherat cement factory)

Quantity	**Value**	**Unit**
Outer diameter	4.4	m
length	52	m
h	8.4	W/m^2K
T_s	300	^oC
T_∞	40	^oC
q_{conv}	1569	KW

To quantify the radiation heat losses from the kiln surface, we use two bodies radiation exchange formula given in the textbook [4]. From the mentioned formula, heat losses can be calculated if we know the emissivity of kiln surface. As material of the kiln shell is mostly mild steel and heavy oxidized, thus assuming an external surface emissivity of 0.9 we can use the formula to calculate radiation heat losses. The calculations are given in table 6.

Table 6: Shows parameters for calculation of radiations heat transfer from kiln surface (Case study: cherat cement factory).

Quantity	**Value**	**Unit**
View factor	1	-
ε	0.9	-
A	718.4	m^2
σ	$5.67 * 10^{-8}$	W/m^2K^4
T_s	300 (573)	℃ (K)
T_∞	40 (313)	℃ (K)
q_{rad}	3600	KW

Now the total heat that has been lost through kiln surface is equal to the summation of the convection heat losses and radiation heat losses. In order to find the change in the outside kiln surface temperature with the change in surface emissivity, ANSYS steady state thermal module was used. After creating the geometry, the ANSYS default mesh was used for meshing and the meshing was done to ensure the maximum skewness less than 0.85. The model was then transferred to steady state thermal for analysis. The inside kiln surface temperature was kept at a boundary condition of 1800 Celsius. The thermal conductivity and the outside convective heat transfer coefficient along with outside ambient temperature was then set as boundary condition. The emissivity was then given and was changed with an increment of 0.05 and the outside kiln shell surface temperature was then noted. Thermal resistance before and after

applying the paint is assumed constant and is calculated from the heat transfer before applying the paint. Now when paint is applied on the kiln surface its surface temperature increases and the gradient for the total heat transfer decreases. Thus, heat losses after paint are reduced which is the difference between these two and is actually the saving of energy.

Results:

Results of combustion analysis of rotary kiln in CFD Based on our model and boundary condition we performed our analysis. The results can be displayed in many ways out of which contours and graphs are the best methods to visualize the results. In order to display the contours of temperature, we drew a plan which intersected the kiln in the middle. The temperature is then displayed on the plan. The contours are shown in the figure 3.

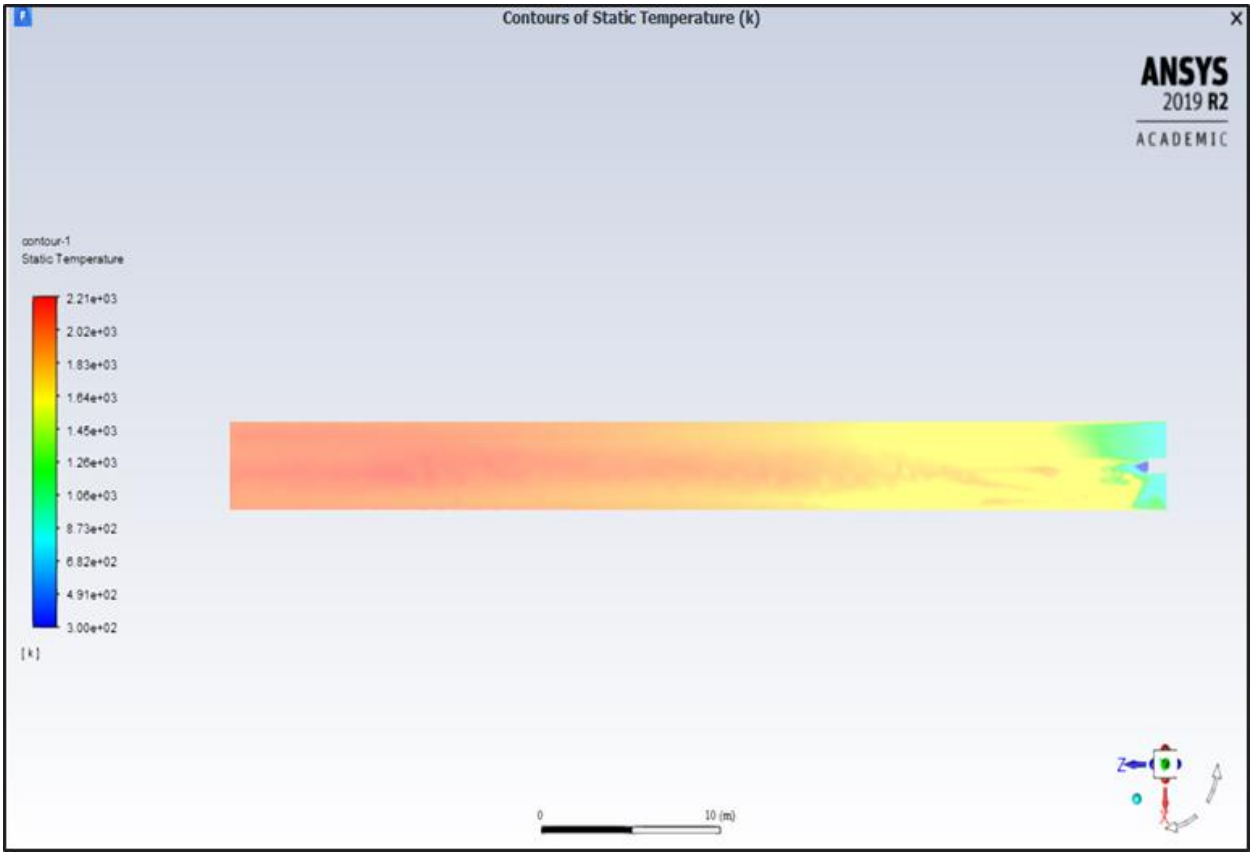


Figure 3:Temperature contour of the gases along the length if the kiln.

This contour which is given above shows the temperature of the gases along the length of the kiln. To study the temperature of kiln interior wall, we drew a line on the wall of the flow volume along the kiln length and plot the temperature profile along the line [6]. To measure the gases temperature in the middle of the kiln, we drew a line in the middle of the kiln form one side to the other side of the kiln and plotted the temperature profile along the mid line too. The graph of the wall temperature (1800 ℃) and gas temperature (2100 ℃) is combined and is displayed below.

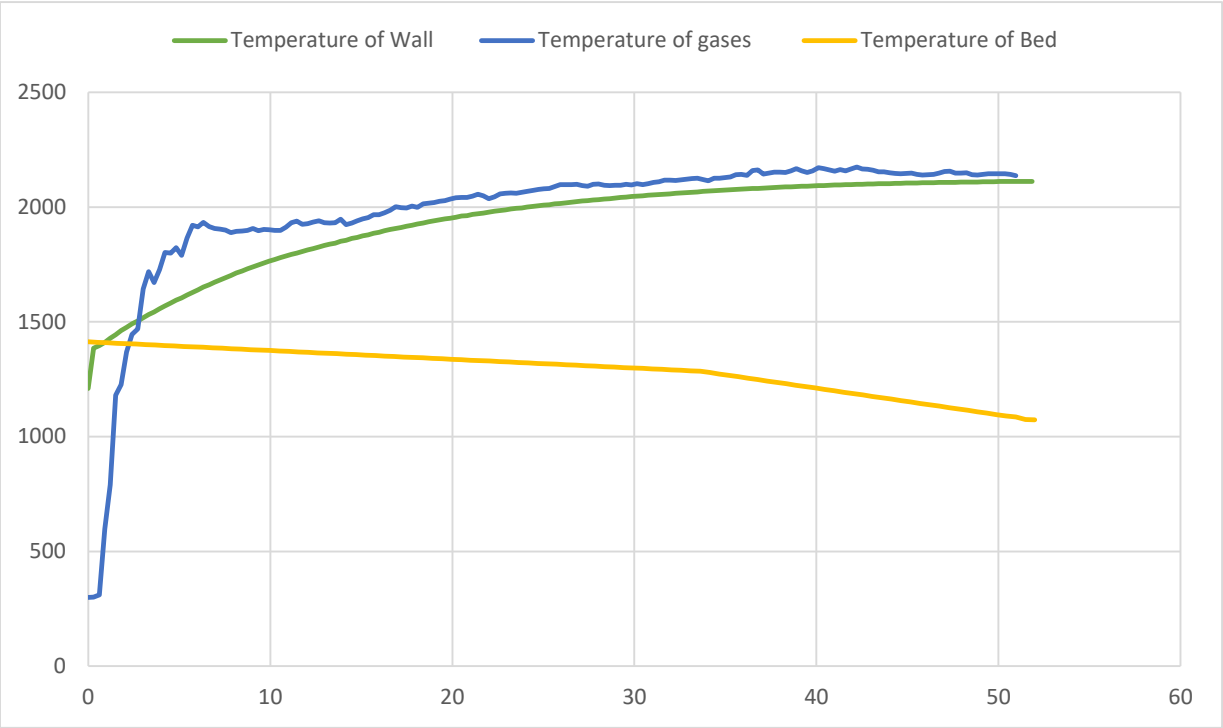


Figure 4: Wall and gases Temperature profile at 1 kg/s of fuel.

It can be seen that the wall temperature is always less than the gas temperature at the corresponding position. The burner tip has relatively low temperatures because the combustion has just started and as we move along the kiln the temperature increases until a certain point.

For further validation of our results, we compared our results with similar analysis in previous literature [8] and our results matched their results. There were no sensors places inside the kiln of Cherat cement factory so we have no data form the factory for comparison. There are zones in the rotary kiln with temperature ranges needed for the calcination process to occur. By dividing the kiln into zones and then extracting the maximum and minimum temperatures, we can find the range obtained from the combustion analysis in CFD. Our results matched after comparing the results with the temperature ranges of zones from the Cherat cement factory kiln.

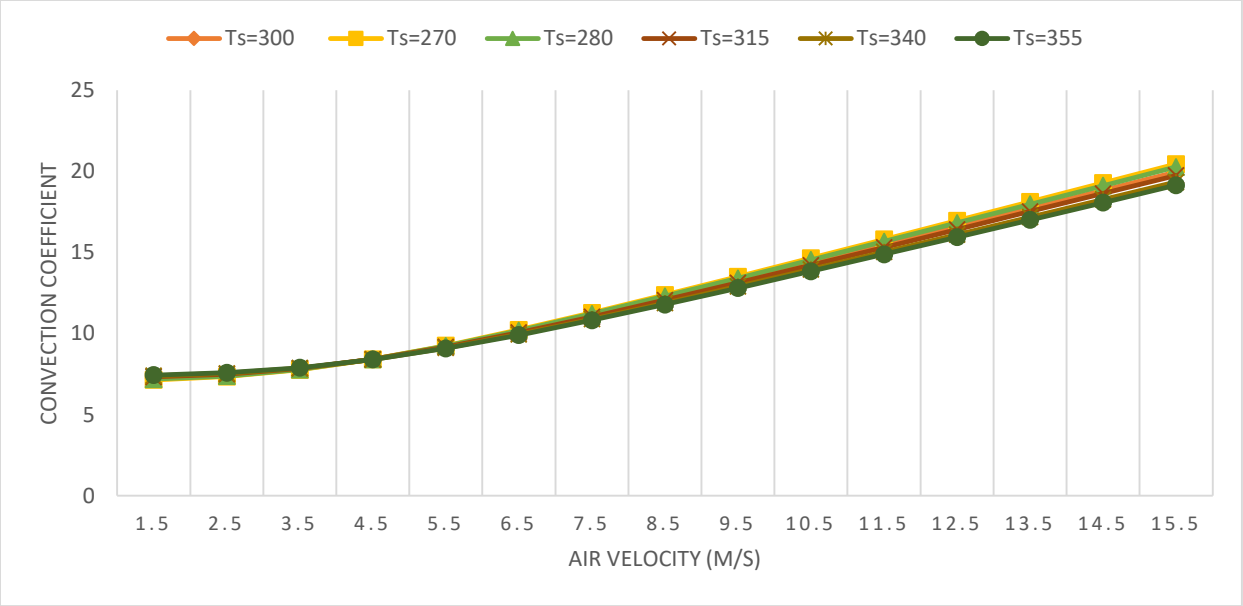


Figure 5: Shows mixed convection coefficient for different velocities at different surface temperatures.

Figure 5 above shows the variation of convection coefficient with different air velocities. The velocity is represented on the horizontal axis while the vertical axis shows convection coefficient. From the figure it can be seen that below the velocity of 5 m/s the convection coefficient is almost constant, which shows natural convection dominates below 5 m/s and beyond that it increases linearly which means that forced convection dominates beyond this point.

Moving forward we know that radiation heat losses depends on surface emissivity and its value decreases as surface emissivity decreases thus the total heat losses from kiln surface depends upon initial surface emissivity of kiln. The variation in radiation heat losses with different surface temperature is in figure below.

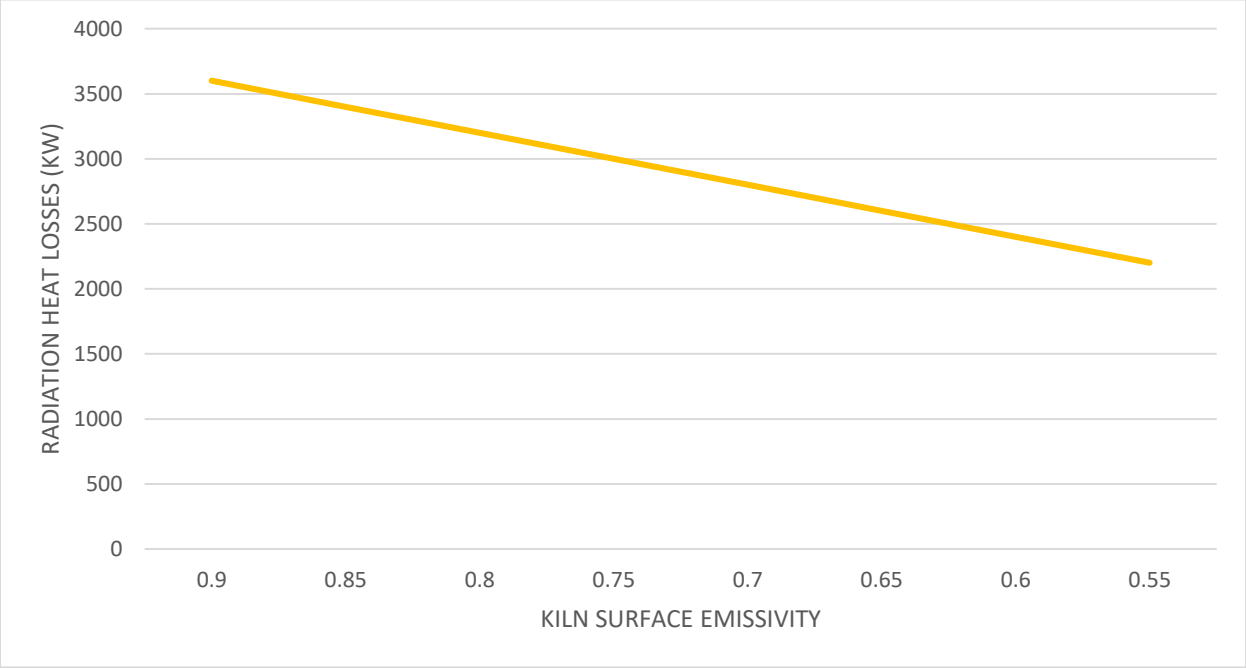


Figure 6: Shows decrease in radiation heat losses with decrease in surface emissivity.

The total heat that has been lost through kiln surface is equal to the summation of the convection heat losses and radiation heat losses.

$$q_{tot} = q_{conv} + q_{rad} \quad (3)$$

The total heat losses from the 52 m long and 4.4 m diameter kiln, whose surface is at an average temperature of 300° C (573K) and 0.9 emissivity are 5173 KW. It can be easily shown that 30% of these losses are due convection and 70% are due to radiation. So, it is concluded that radiation heat losses are almost twice stronger than convection heat losses. Total heat losses depend upon surface temperature, kiln dimensions and its surface area. The sensitivity of total heat losses, convection heat losses and radiation heat losses are shown in figure 7.

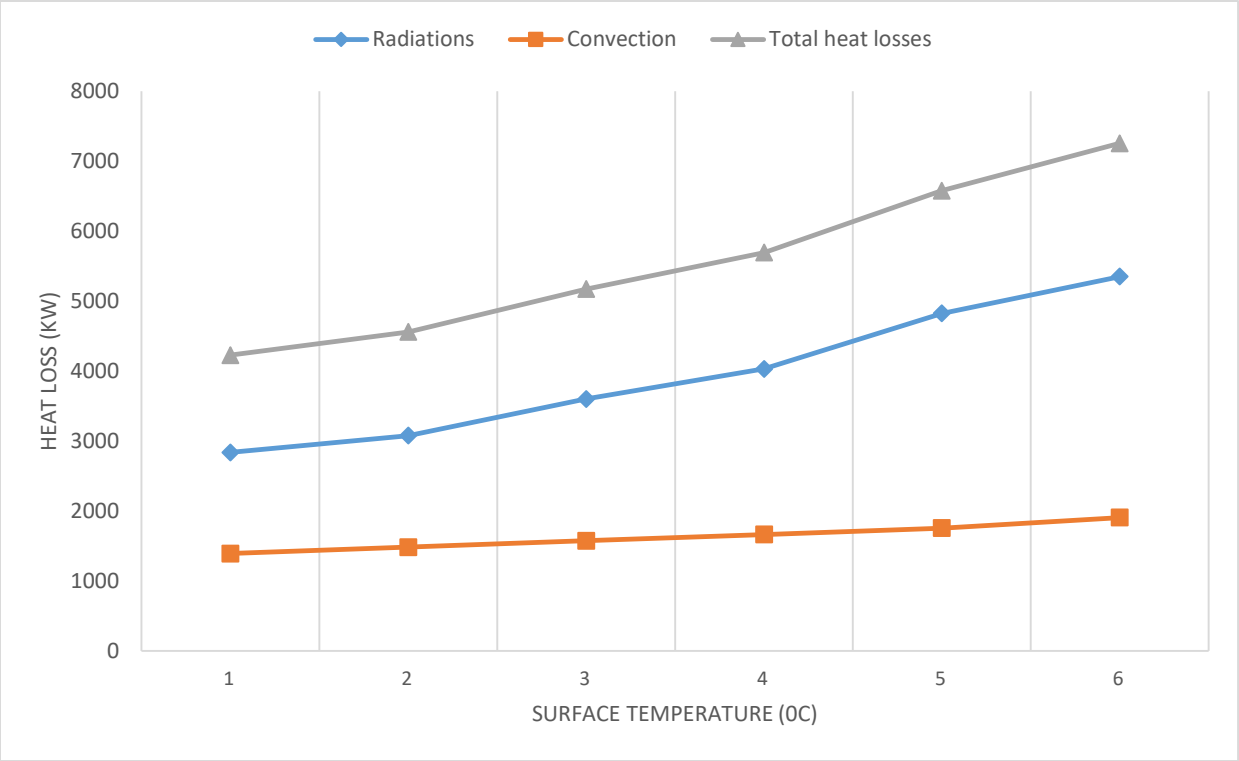


Figure 7: Shows variation of heat losses for different kiln surface temperature.

From figure 7, it can be seen that total heat losses increase as temperature increases. Taking heat losses at 300 Celsius as reference, the percent increase or decrease in total heat losses is in the table 7.

Table 7: Shows percent increase or decrease in heat losses at different surface temperature.

Temperature (^{0}C)	**Losses (KW)**	**% age**
270	4226	-18
285	4558	-12
300	5173	Reference
315	5694	+10
330	6594	+27

For a surface temperature of 315 oC, total heat losses increase by 10%. Table 4-7 summarizes heat losses for various surface temperatures and the percent increase or decrease in total heat losses with reference to 300 Celsius.

it can be seen that decreasing surface emissivity from 0.9 to 0.85 the surface temperature increases by 6 Celsius. Based on various references and industrial data an increase in surface temperature up to 50 Celsius is safe for maintaining kiln functionality. Thus suggesting a surface emissivity of 0.6 because of which the surface temperature will increase by 45 Celsius.

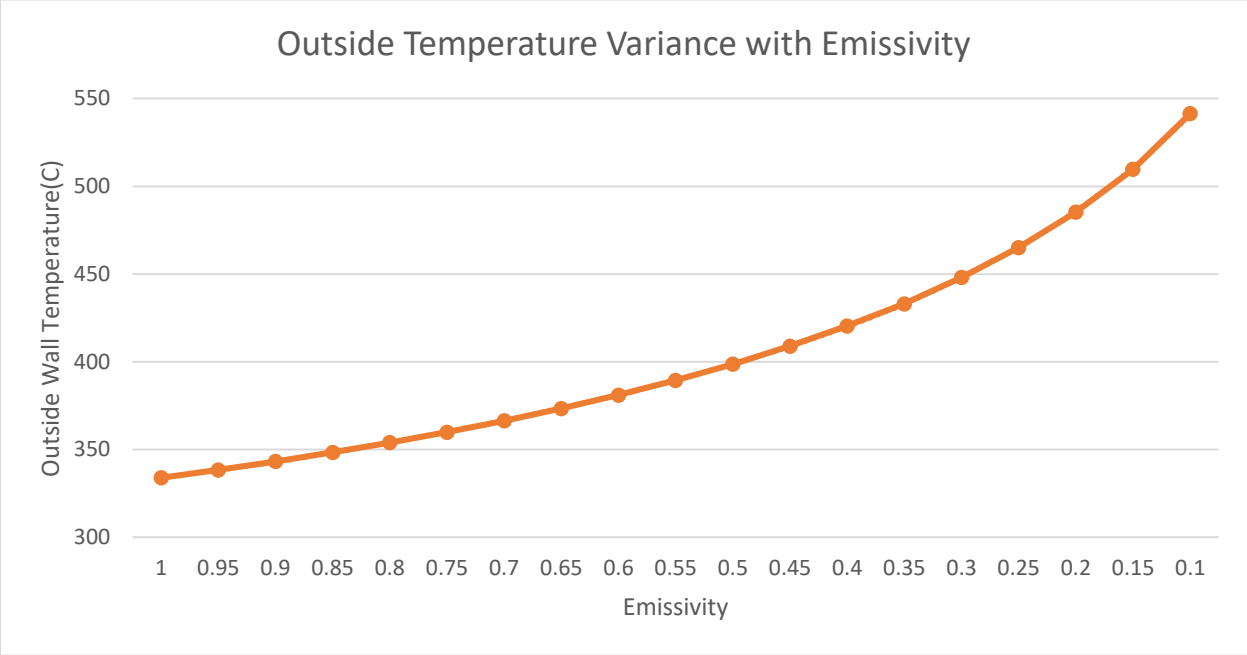


Figure 8: shows increase in surface temperature when emissivity decreased.

For 0.6 emissivity paint, the saving in energy is 250KW which is 4.8% of the total heat loss if we consider a 4.4-meter internal diameter kiln with 52-meter kiln length and a surface temperature of 300 Celsius.

Discussion:

A substantial amount of energy is lost due to radiation and convection heat losses from the rotary kiln. The cement industry is one of the leading sector which uses tremendous amount of energy and most of it is used for calcination process which happens in a rotary kiln. Now our proposed method uses a low emissivity paint which reduces the radiation heat losses and is relatively easy to implement considering the alternatives for energy savings. Our method focuses on applying a thin sheet of paint that will reduce the overall emissivity of the kiln surface thus adding in blocking radiation heat loss.

There is also a concern that needs to be addressed before lowering the emissivity and that is ensuring that the structural integrity of the kiln remains stable. We have seen that high thermal stresses are induced if a radiation heat losses are reduced. Blocking radiation heat losses lead to increase in surface temperature which results in high thermal stresses that can damage the rotary kiln. In order to avoid this issue, a critical value must be found that addresses the concern while also achieving the goal of reducing heat loss.

Our study finds out the total heat losses from the kiln by assuming a mixed convection model. The total heat loss is the combination of radiation as well as convection heat losses. We also found out the critical emissivity value and found out that with each 0.05 reduction in emissivity, we get an average increase of 6 Celsius in surface temperature. Previous literature suggests that if we keep the keep the surface temperature below 350 Celsius, we can address the thermal stresses problem. In our case we suggest a value of 0.6 emissivity which increases the surface temperature by 45 Celsius while also giving us a total saving of 250 kilo watt. Based on our coal composition we get a low calorific value of 2.81e 07 j / kg . Now if we use the calorific value of the coal and convert it into fuel saved on a daily basis,

that will turn out to be About 890 kg/day. So in conclusion a huge amount of coal is saved by painting the surface with 0.6 emissivity paint and the carbon emissions decrease by almost 3.5 tons annually.

Our proposed method not only saves money in terms of fuel cost but also reduces the carbon footprint with a reduction of 3.5 tons annually. This will have a huge impact on environment which was our secondary goal.

Conclusion:

For a typical kiln geometry (external diameter = 4.4m & length = 52m) whose average surface temperature is 300 Celsius, total heat losses by mixed convection model and thermal radiations are 5170 KW, 30% of the total losses are due to surface convection while 70% are due to thermal radiations thus radiation losses are twice as strong as convection losses. Also type of convection from kiln surface depends upon its geographic location. Naturel convection is appropriate when average air velocity is less than 4 m/s, forced convection is appropriate when average air velocity over the kiln is greater than 5 m/s and mixed convection is best when average air velocity is in between 4-5 m/s.

Based on the detailed heat transfer inside the kiln it is concluded that painting the surface with a paint of emissivity 0.6 will decrease the total heat losses to 4920 KW, convection losses will increase while radiation losses will decrease thus net effect is saving of 250 KW energy. As result, the surface temperature will rise by 45 Celsius while also giving us a saving of 3.5 tons of carbon annually in terms of fuel.

Paint of lower emissivity will save greater amount of energy but will increase temperature as shown in figure 8, which can cause structural problems in kiln. A large increase in kiln surface temperature is not desired because it will cause structural problems induced by large differential thermal expansion between metal shell and refractory layer. Based on case study an increase in surface temperature up to 50 Celsius will not affect the function of the kiln therefore the paint that will increase shell temperature up to 50 Celsius is appropriate.

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